

Machine Learning Algorithms for Load Forecasting Based on Big Data

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Abstract—Big Data's importance in the context of modern Power Systems' transition is rapidly increasing. Thus, the research interest towards techniques to explore and extract the most relevant information is in a continuous expansion. In this framework, Machine Learning algorithms are the most suitable approach towards retrieving valuable knowledge in a heterogeneous and ever more complex data environment. Considering the high rate and velocity of data production in smart Power Systems, Machine Learning implementation is mandatory. Load forecasting is a hot topic due to its high fluctuation and economic impact considering the increasing and variable Renewable Energy penetration. This paper presents a comparison between the statistical load forecast and two Machine Learning algorithms based on multiple types of errors. It highlighted the necessity of refining the algorithms and enhancing the available data using Feature Engineering to improve the performance. Nevertheless, Machine Learning algorithms are the only approach feasible for the ever-increasing data pool with heterogeneous characteristics.

Keywords—Big Data, Decision Tree, Linear Regression, Load Forecast, Machine Learning

I. INTRODUCTION

This article aims to present the main data generating points in a Power System and to calculate the load forecast using different Machine Learning (ML) algorithms in order to minimize the errors when compared to the statistical forecast used currently by ENTSO-E (European Network of Transmission System Operators). Furthermore, the paper will emphasize the importance of Big Data in modern Power Systems.

Big Data cannot be currently easily defined. A consensus has been reached between the emerging descriptions: this is an up and coming technical problem due to sets of high-volume heterogeneous data, that require new algorithms and frameworks in order to obtain the most useful information. Big Data definitions are therefore subject to the data mining algorithms evolution and the corresponding hardware upgrades necessary for high quantities of data. Defining this concept is still in a relative state, an absolute definition being impossible at this time. Accepted definitions are based on four main ideas, as follows:

- Definitions that rely on the Big Data characteristics and that have in the foreground the "V" aspects: Volume, Variety, Velocity, Value, Veracity [1].
- Definitions that focus on the fact that traditional data management methods and traditional relational databases are not sufficient for Big Data

and explain the techniques and technologies that are needed/useful, such as advanced analytical techniques [1].

- Definitions that relate more to the purpose and reasons why we use Big Data (analytical).
- A combination of the above and others.

Data analysis currently plays a vital role in technical systems nowadays. The conventional electricity networks are required to gather, deposit and analyze data coming from smart devices in the grid. This additional layer of information is driven by the unfolding of the communication and information technology. The present paper aims to describe Big Data with all its characteristics, Smart Grids and the enormous quantities of data generated by them to highlight the potential benefits of Advanced Data Analytics in the power system sector [2]. It also addresses the basic concepts and common procedures for data analysis for general problems. By integrating a huge amount of data into the electricity grid (such as the weather data, geo-location data) a number of advantages can be introduced to the electricity supply service and the quality of life and social well-being in the information age can and will be improved. However, in order to develop the use-cases of Big Data analytics in actual smart grids, multiple problems need to be addressed in a realistic and innovative manner, such as processing techniques, social awareness and the synergies between them [3].

The main steps of the methodology used in this article have been the following:

- Analysis of the state-of-the-art papers in the Power Systems Big Data utilization area.
- Identifying the main flow of Big Data in a Power System and data sources.
- Defining the comparison characteristics and errors that describe the algorithms' performance.
- Applying Machine Learning algorithms to forecast load and comparing it to the statistical results of ENTSO-E.

As can be seen in Figure 1, in the energy field, the concept of Big Data is represented in numerous subdomains, being in full development. The subfields where Big Data is of the greatest interest so far are represented by Energy Storage and Renewable Energy. At the same time, the use of this concept in the Maintenance of electrical substations is not yet in the top of scientific interests.

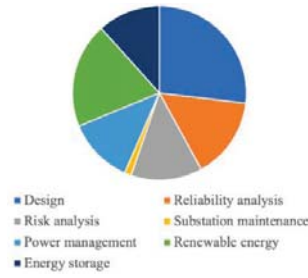


Figure 1. Representing Big Data in the current energy science field

Figure 2 shows the evolution over time of the interest of the scientific community in the energy field towards the concept of Big Data. It can be noted that the implementation of this concept began around 2010, and with the year 2014, an exponential increase in the number of articles on this topic can be observed. However, some subfields such as Maintenance of substations require an intensification of scientific research into the concept of Big Data.

Figure 2 was obtained by querying and processing data about indexed articles using Scientific Data Bases (SCOPUS) using publishing years and domains criteria. This figure proves the increasing interest in the applicability of Big Data in the energy field. Furthermore, it accentuates the potential of some yet unexplored subdomains (e.g., Substation maintenance).

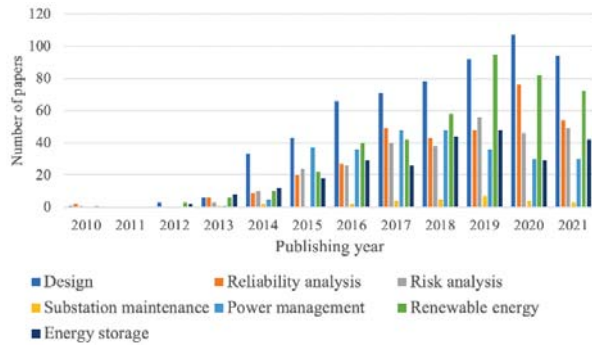


Figure 2. Evolution of the number of articles on Big Data in the energy field - Distribution by subfields

Based on these outstanding results in multiple sectors, Big Data analysis in modern intelligent energy systems provides powerful use-cases. Big Data might empower dependable, effective, and profitable operation through advanced computing technology. Thus, improved forecasting can be obtained through the usage of data analytics applications in energy production and consumption, intelligent meter analysis, asset management and analysis, stable network configurations, cost-efficient energy trading, customer satisfaction, energy efficiency, marketing, detecting invalid data, detecting energy theft, managing, forecasting tasks, and managing outages [4].

This paper focuses in analyzing the data sources in the modern Power Systems and proposes two Machine Learning algorithms that predict load forecast based on Big Data in the following sections as described forward. Section II presents the currently used data sources. Section III describes the methodology for error calculation used to determine the ML algorithms performance. Section IV is the final section of this article, and it presents the main conclusions and future developments in the analyzed area.

II. DATA SOURCES

As an intelligent system comprising the energy subsector as well as the information one, the intelligent network represents a rich source of information, covering all the stages of energy process flow from production to end-user consumption. It includes electrical data from transmission substations, distribution substations, smart meters and non-economic data such as end-user, weather and regional data [5]. These provide a fundamental help in the planning and control of power plants, the maintenance of essential electrical equipment and the analysis of business behavior. [6]. The flow of Big Data in Power Systems is represented in Figure 3, and it shows the following steps:

- **Data Acquisition** – represents the Data Sources of any Big Data flow and it receives data from equipment, weather stations, smart meters. The data is either stored and sent in bulk to the next step or it is a real-time stream for live applications.
- **Data Preparation** – in this step the data is cleaned, homogenized, and structured in a way that will make it easier to be integrated in algorithms.
- **Big Data Analysis** – this step consists in integrating algorithms capable of using large amounts of data (like Machine Learning) and obtaining results and predictions for the variables of interest.
- **Decision Making** – based on the information provided by Big Data Analysis, in this step the stakeholders can provide actionable tasks that will interact directly with the Power Systems.
- **Practical Applications** – the results and decisions can be integrated in different stages of a Power Systems project (e.g., design, operation, and maintenance)

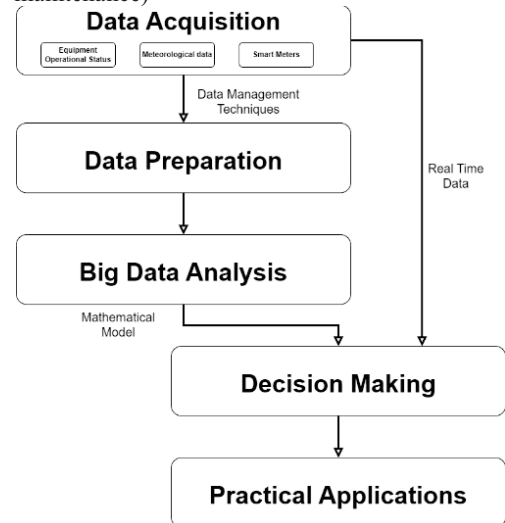


Figure 3. Flow diagram of Big Data type in a Power System

Power systems are characterized by different datasets that are generated as follows:

A. Equipment data

The results of equipment data collection are so widespread and may contain data from transformers, switching equipment, overhead and cable transmission lines and substations [7].

B. Distributed generation data

Each micro-renewable energy plant may produce data: Wind power plants, Photoelectric power plants, Biomass plants.

C. Energy storage systems data

The energy storage management system produces data regarding voltage at the terminals, state of charge (SoC), thermal conditions, fault detection data, operating parameters.

D. Electric networks loading data

The load level of electrical equipment is a comprehensive source of large volumes of data. Big data can provide insight into end-user behavior such as load patterns. Consumers can also share information about the consumption reduction they have achieved.

E. Environmental data

Numerical weather forecast data is a large volume of data transmitted at various speeds in various structures (standard or timestamp). It should be noted that weather prediction equations are extremely complex and non-linear.

F. Sensor networks

These networks are comprised of: Phasor Measurement Unit (PMU), wireless connections, cloud storage, Internet of Things (IoT), specialized monitoring sensors, data sources outside the power field and public datasets.

For this article, the main datasets used were extracted directly from ENTSO-E using the official API.

ENTSO-E comprises 39 electricity transmission system operators (TSOs) across 35 countries in Europe. It includes EU and non-EU countries and was established receiving legal mandates through the EU's third package for the internal energy market in 2009, which aims to further liberalize the EU gas and electricity markets [8].

The load data and forecast data was received by ENTSO-E from the National Electric Energy Transport Company "Transelectrica". The published information had to comply with the regulations specified by ENTSO-E [9].

The input consists of datasets comprising the load and the forecasted load in float format per hour over the analysis period.

In this regard, ENTSO-E has developed an Application Programming Interface (API) through which, upon request, the data collected on this platform can be accessed.

The API is a software interface dedicated to enabling user-free interaction between two applications. API consists of an assortment of software functions and procedures. API represents a software code that can be accessed or executed and that helps two different applications communicate and exchange data between them [10].

ENTSO-E was chosen as baseline because it is the official load forecasting database in use by all the stakeholders in the national power system.

In this document, the following characteristics are used for the ENTSO-E API datasets:

- Country of analysis: Romania
- Analysis period: 01.01.2017 – 31.12.2021
- Countries for energy exchange: Bulgaria, Serbia, Hungary, Ukraine, Republic of Moldova.

III. ERRORS CALCULATION

Regarding forecasts extracted by API ENTSO-E, the following errors will be used to calculate the deviation of the forecasted values from the recorded ones.

Mean Square Error (MSE) – represents the square error prediction according to Equation 1 [11].

$$MSE = \frac{1}{m} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

where:

m – represents the number of elements in the entire test process (=1 for all values considered)

y – the actual value of the item tracked for error

$\hat{y}$ – the prediction value of the item tracked for error

n – number of real values / predicted

Root Mean Square Error (RMSE) – it is a performance metrics deducted from the MSE value using Equation 2 [12].

$$RMSE = \sqrt{MSE} \quad (2)$$

This error has calculated values close to the estimated target values and provides easy-to-understand information about the impact of the error on the forecasted values.

Mean Absolute Error (MAE) – Units of error value match units of estimated target value. Unlike RMSE, MAE exhibits a linear response to changes, making it more intuitive. MSE and RMSE amplify the impact of large errors due to the squaring of error values, which inflates the average error. In contrast, MAE treats different errors equally and scores them linearly as errors increase. The MAE score is computed as the average of absolute error values, where the absolute function ensures that the difference between expected and predicted values is consistently positive. This guarantees a positive outcome when calculating MAE, regardless of the direction of errors. This error is calculated using Equation 3 [13].

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

Mean Absolute Percentage Error (MAPE) – or Mean Absolute Percentage Deviation (MAPD), serves as a statistical measure to assess the accuracy of a forecasting method [14]. Typically, it quantifies accuracy by using the following formula as a ration as follows:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_t - F_t}{A_t} \right| \quad (4)$$

where: A_t is the actual value and F_t is the forecast value.

For the test set, the errors calculated for the ENTSO-E forecasts are shown in Table 1. Table

TABLE 1. CALCULATED ERRORS FOR ENTSO-E LOAD FORECASTS

Calculated error type	Calculated error value
MSE	97524
RMSE	312
MAE	234

IV. LOAD FORECASTS USING MACHINE LEARNING ALGORITHMS

Usually, ML algorithms tackle either a regression problem with an expected continuous variable or a classification problem with a categorical variable as a result. In this case, a regression problem was identified [15].

The algorithms presented in this paper show the power of ML even with a generalist approach. Further research would show more specialized load forecasting algorithms, but, nevertheless, the algorithms presented in this paper reach satisfactory results without high demands on the computing processes. The transmission network is the target of the calculation. The algorithms were developed in Python 3.10.6 using the *Jupyter Notebook* extension. The main libraries used are Pandas and Scikit-Learn.

A. Linear Regression

Linear regression [16] is one of the supervised machine learning algorithms that observe continuous characteristics and predict a result. The number of input variables determines the following types of linear regression:

- Simple linear regression – if a single input variable is used.
- Multiple linear regression – if more than one input variable is used.

This analytical approach calculates the coefficients for a linear equation, which incorporates one or more independent variables to effectively predict the dependent variable's value. Linear regression aims to establish a straight line or surface that reduces the disparities between predicted and actual output values. In the case of this paper, a multiple linear regression is therefore used.

The results obtained for the analysis period are displayed in Figure 4.

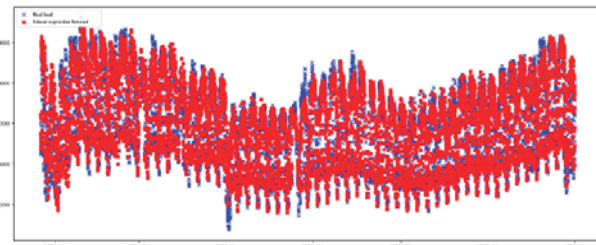


Figure 4. Load forecast results – Linear regression

For the set of test data considered, in the case of linear regression, the calculation errors shown in Table 2 were obtained:

TABLE 2. CALCULATED ERRORS FOR LOAD FORECASTS – LINEAR REGRESSION

Calculated error type	Calculated error value
MSE	110211
RMSE	332
MAE	237
MAPE	3.51%

The results of the linear regression load forecast present slightly bigger errors than the algorithm used by ENTSO-E. This is most likely due to the generalization of the meteorological data and the lack of more specific inputs.

B. Decision Trees

A decision tree [17] serves as a decision making aid employing a tree-like structure to depict decisions, their potentials outcomes and factors like random events, resource costs and utility. It provides a visual representation of an algorithm, primarily consisting of conditional control instructions. A decision tree is a supervised, non-parametric learning algorithm. It is characterized by a hierarchical tree structure that includes a root node, branches, internal nodes and leaf nodes. The results obtained for the analysis period are presented in Figure 5.

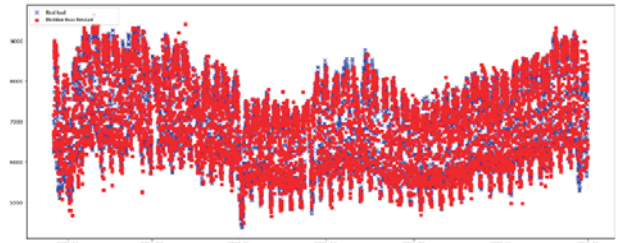


Figure 5. Load forecast results – Decision trees

For the test dataset considered, for the decision trees, the calculation errors given in Table 3 were obtained:

TABLE 3. CALCULATED ERRORS FOR LOAD FORECASTS – DECISION TREES

Calculated error type	Calculated error value
MSE	109486
RMSE	330
MAE	220
MAPE	3.4%

The errors obtained by applying the decision tree algorithm are lower compared to the linear regression algorithm. When compared to the linear regression in terms of errors, decision trees show a: 0.7 % decrease of MSE, 0.6 % decrease of RMSE, 7.2 % decrease of MAE, 0.11 % decrease of MAPE.

This is a consequence of implementing a more complex and advanced logic. However, the performance of this algorithm is still lower than the ENTSO-E algorithm.

V. CONCLUSION AND FUTURE DEVELOPMENT

This research is a starting point for future research that will address more specialized ML load forecasting algorithms. The processing of large datasets is the most important step in using Big Data for the purposes of improving electrical power calculation. It is necessary to

develop increasingly complex algorithms that are based on these large datasets.

The main stakeholder for the information in the case study is represented by the energy dispatchers (both national and territorial). These dispatchers can use the improved information from the load forecasts to make decisions underlying the stability of the energy system. Another interested party in such forecasts is represented by legal entities acting in the energy market. A better load forecast can lead to more profitable business decisions that favor the participants in the energy market (e.g., for the reserve capacities of the power system) [18].

The research presented in this article proved that basic ML algorithms do not provide better results in terms of errors compared to the statistical forecasts provided by ENTSO-E. However, it is important to point out that, even though the calculated error values are higher for the analyzed algorithms, they are quite close to the very complex statistical one. Nevertheless, the statistical approach has its limitations (e.g. it is slow to react to big changes in the data used). These limitations can be overcome by better refined ML algorithms. By using Feature Engineering, a lot more data is taken into consideration to bring the mathematical model closer to the real physical phenomena (thus improving the ML way further than the possibilities of statistical forecasting). Machine Learning algorithms are better suited for Power Systems because they can ingest and process Big Data in a way more efficient manner and will represent the most adequate approach to future challenges faced by Power Systems.

Future developments will include more advanced load forecast algorithms and generation forecast calculations using similar methodologies as the ones presented in this article. This domain is as important as that of load forecasts and together may provide a significantly enhanced description of the Power System. Further, load forecasts for smaller regions would be a focus, being expected that with the regionalization of calculations, the accuracy of forecasts will increase. Moreover, feature engineering will probably improve the forecast results (such as weather, geographic position, and others).

ACKNOWLEDGMENT

For the elaboration of this paper, we would like to thank ENTSO-E for facilitating access to the country-level databases they own, and which have been used as a basis for the work in question.

The results presented in this article have been funded by the Ministry of Investments and European Projects through the Human Capital Sectoral Operational Program 2014-2020, Contract no. 62461/03.06.2022, SMIS code 153735.

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